

SLAM Algorithm Selection and Pareto Ablation

Standalone draft for review

August 4, 2026

Reviewer note. Every number in the tables and figures below is computed directly from the recorded bags by `PAPER_IMAGES_HIL/generate.py`; none is typed by hand. Re-running that script after further trials updates this document. Known gaps are listed at the end of `README.md` and are deliberately left as gaps rather than filled with estimates: the offline EuRoC results are an outline only, ORB-SLAM3 has no front-end latency figure, and the OV2SLAM fast profile is scored over a shorter path than the other arms because it does not initialise within the initialisation leg.

0.1 SLAM Algorithm Selection and Pareto Ablation

0.1.1 Offline evaluation on EuRoC playback

Setup. Candidate backends were screened on EuRoC MAV playback across ten sequences (MH_01 to MH_05, V1, V2), ten runs per sequence, replayed through an in-process rosbag adapter for deterministic frame delivery. This stage selects which systems are worth the cost of hardware-in-the-loop trials.

Candidates.

- **ORB-SLAM2:** feature-based monocular SLAM with loop closure; the accuracy reference for this class.
- **ORB-SLAM3:** successor with an Atlas multi-map backend and improved relocalisation.
- **OV2SLAM-accurate:** optical-flow front end tuned for accuracy (single-scale detector, CLAHE enabled).
- **OV2SLAM-fast:** same codebase, throughput-tuned detector (FAST corners, multi-scale, no CLAHE).
- **RTAB-Map:** appearance-based graph SLAM, included offline only; its odometry front ends are RGB-D, stereo and ICP, with no monocular node, so it cannot enter the monocular sweep of Section 0.1.2.

Metrics. Monocular trajectories are recovered up to scale, so alignment solves for scale:

$$(s^*, \mathbf{R}^*, \mathbf{t}^*) = \arg \min_{s>0, \mathbf{R} \in SO(3), \mathbf{t}} \sum_{i=1}^N \|\mathbf{p}_i - (s\mathbf{R}\mathbf{q}_i + \mathbf{t})\|^2. \quad (1)$$

$$\text{APE}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|(s^*\mathbf{R}^*\mathbf{q}_i + \mathbf{t}^*) - \mathbf{p}_i\|^2} \quad (2)$$

Accuracy is reported as the median of APE_{RMSE} over runs, in metres. Reliability is

$$R = 100 \frac{N_{\text{completed}}}{N_{\text{attempted}}} [\%], \quad (3)$$

counting a run as completed only if it tracked to the end of the sequence and produced a trajectory.

Table 1: Offline evaluation on EuRoC playback. RTAB-Map is reported over ten sequences from the best-run selection table; all others over 100 runs.

Candidate	runs	reliability [%]	median APE [m]	latency [ms]	CPU [%]
ORB-SLAM2	100	100.0	0.0545	39.5	164.1
ORB-SLAM3	100	74.0	0.0546	32.5	147.9
OV2SLAM-acc	100	100.0	0.1271	n/a	70.0
OV2SLAM-fast	100	100.0	0.2974	n/a	38.7
RTAB-Map	10	100.0	0.1049	n/a	n/a

Table 2: Online sweep in the simulation environment, ten trials per candidate.

Candidate	runs	reliability [%]	median APE [m]	latency [ms]	CPU [%]
ORB-SLAM2	10	100.0	0.3569	28.4	80.5
ORB-SLAM3	10	100.0	0.3162	31.1	74.4
OV2SLAM-acc	10	100.0	0.4166	8.5	37.7
OV2SLAM-fast	10	100.0	0.4560	4.6	33.9

Results. Table 1 reports the offline comparison; Figure 1 (left) and Figure 2 show accuracy and reliability. ORB-SLAM2 and ORB-SLAM3 are equivalent on accuracy (0.0545 m and 0.0546 m median APE), but ORB-SLAM3 completed only 74% of attempted runs against 100% for every other candidate. The OV2SLAM profiles trade accuracy for cost in the expected direction, the fast profile losing a factor of 2.3 in accuracy relative to the accurate profile while halving CPU. RTAB-Map sits between them at 0.1049 m.

0.1.2 Online sweep in the simulation environment

Setup. The shortlisted monocular backends were flown in the hardware-in-the-loop simulator on a fixed approach trajectory, ten trials each, with the environment, oracle detector and controller held constant so that the SLAM backend is the only manipulated variable. SLAM is recorded and evaluated but is not in the control loop, so every arm flies the same path.

Candidates. The four monocular systems of Section 0.1.1: ORB-SLAM2, ORB-SLAM3, OV2SLAM-accurate and OV2SLAM-fast. RTAB-Map is excluded for the reason given above.

Metrics. Accuracy and reliability as defined by (2) and (3), scored over an identical window for every arm (mission start to arrival). Cost is characterised by front-end latency per frame and mean CPU utilisation on the target platform. Pairwise differences in median APE are assessed by an exact permutation test with Holm correction, reported in Table 3.

Results. Table 2 reports the sweep. All four backends completed 100% of trials. Median APE spans 0.316 m to 0.456 m, a spread of 44%, while front-end latency spans 4.6 ms to 31.1 ms and CPU utilisation 33.9% to 80.5%.

Pareto discussion. The two axes separate very differently. Accuracy varies by 44% across the four backends, and the separation is only partly significant: the ORB-SLAM pair are indistinguishable from each other, as are the two OV2SLAM profiles, but every ORB-SLAM/OV2SLAM pairing differs (Table 3). Cost varies by a factor of 6.8 in latency and 2.4 in CPU. The cost spread is an order of magnitude larger than the accuracy spread, so on a compute-constrained platform

Table 3: Pairwise differences in median APE, permutation test with Holm correction.

A	B	Δ median APE [m]	p_{Holm}	
ORB-SLAM2	ORB-SLAM3	+0.0406	0.1720	n.s.
ORB-SLAM2	OV2SLAM-acc	−0.0597	0.0388	differ
ORB-SLAM2	OV2SLAM-fast	−0.0992	0.0095	differ
ORB-SLAM3	OV2SLAM-acc	−0.1004	0.0388	differ
ORB-SLAM3	OV2SLAM-fast	−0.1398	0.0180	differ
OV2SLAM-acc	OV2SLAM-fast	−0.0394	0.1720	n.s.

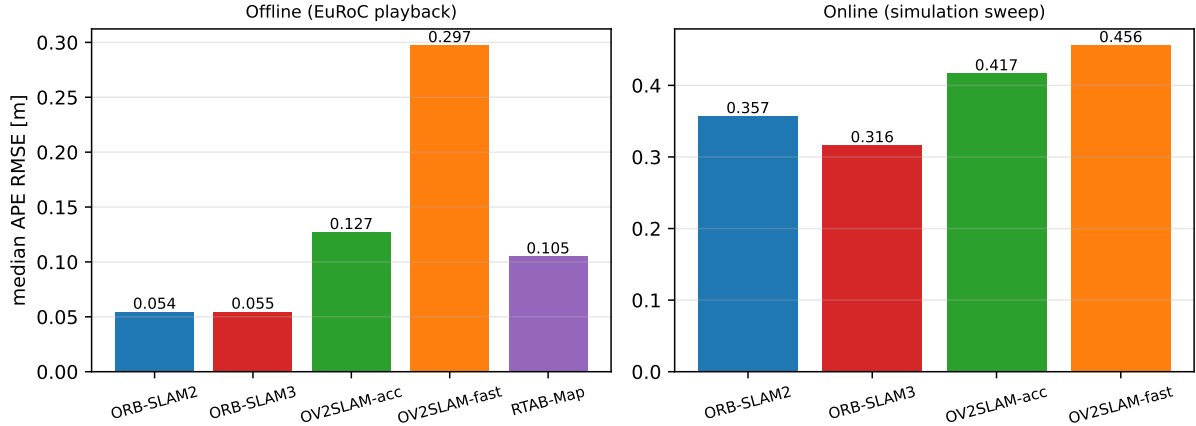


Figure 1: Median APE RMSE by candidate, offline (left) and online (right).

the operative question is not which backend is most accurate but how little accuracy a given compute budget must give up.

Figure 4 makes the non-dominated set explicit. ORB-SLAM3 attains the best median APE, 0.316 m, at 31.1 ms and 74.4% CPU. OV2SLAM-fast attains the lowest cost, 4.6 ms and 33.9% CPU, for 44% more error. OV2SLAM-accurate sits between them, giving up 32% accuracy against ORB-SLAM3 for a $3.7\times$ latency reduction and half the CPU, which makes it the practical choice wherever the perception budget is shared with a controller and a detector. ORB-SLAM2 is nearly dominated: ORB-SLAM3 is both more accurate and cheaper in CPU, and ORB-SLAM2 remains on the front only through a 2.7 ms latency advantage that is unlikely to be decisive at a 13 Hz frame rate.

Two results concern configuration rather than algorithm choice. The two OV2SLAM profiles differ only in feature-detector settings, yet that produces a $2.3\times$ accuracy gap offline, comparable to the difference between distinct systems, even though the same pair is not separable online. Second, the regimes do not agree. Absolute error rises by roughly an order of magnitude in simulation for every backend, the accuracy gap between the ORB-SLAM family and OV2SLAM narrows from $2.3\times$ to $1.3\times$, and ORB-SLAM3’s 74% offline reliability does not reappear in simulation, where it completed all ten trials. Backend selection therefore cannot be transferred from public-benchmark rankings to a specific deployment without measuring in that deployment.

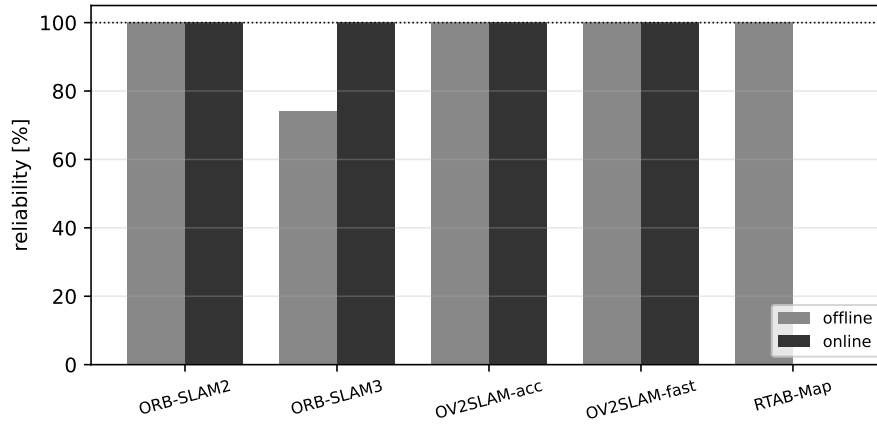


Figure 2: Reliability, the fraction of attempted runs that completed and tracked.

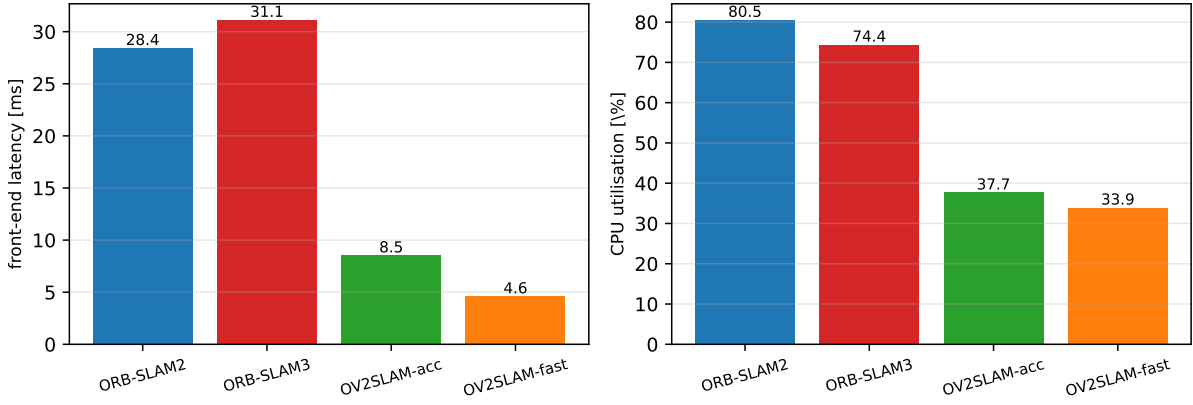


Figure 3: Cost profile in the online sweep: front-end latency per frame (left) and CPU utilisation (right).

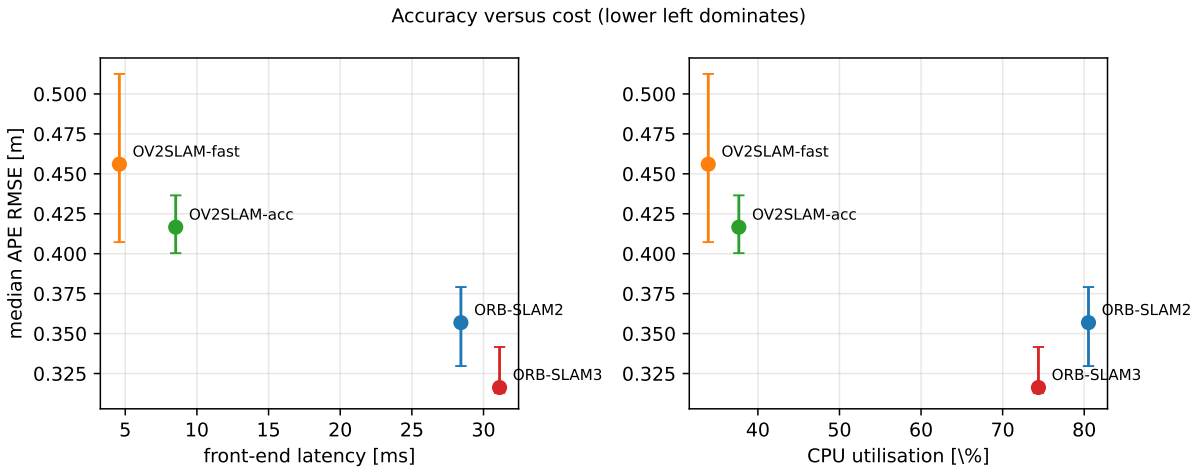


Figure 4: Accuracy versus cost. Points toward the lower left dominate. Bars give the interquartile range of APE over ten trials.